

Badre Mhiouah

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Quantitative Finance, Data Science & Applied AI

EXPERIENCE

Societe Generale Corporate and Investment Banking : Data Scientist

Jun.2025 – Jul.2026

General Inspection

Hong Kong

- Partnered with **non-technical stakeholders** to identify **time-consuming, repetitive audit activities** and assess opportunities for automation, translating business needs into **data and AI use cases**.
- Designed and prototyped **LLM-based solutions** using **prompt engineering and LLM APIs** to automate and accelerate audit workflows, including the enrichment, structuring, and analysis of **operational loss data**.
- Developed a **Power BI platform** combining automated insights and **multi-level risk reporting**, enabling auditors to reduce manual analysis and gain faster access to **actionable risk information**.
- Analyzed **large-scale trading datasets** from Delta One activities to identify **anomalous patterns, risk concentrations, and control weaknesses** across equity and index derivative products, supporting auditors in risk-focused investigations.
- Performed **forensic analyses of trading and market data** to identify potential indicators of **market manipulation**.
- Delivered **Python and Power BI training** to auditors, enabling **self-service analytics** and facilitating the adoption of data-driven approaches within the IGAD teams.

Barclays : Trader intern

Aug.2023 – Jan. 2024

Cross-asset derivatives trading desk

Paris & London

- Developed a **trading strategy aimed at hedging own credit risk**, resulting in a **7% boost in profitability** and a **5% decrease in P&L volatility**. (Bloomberg, Python, Machine learning)
- Improved **risk management instruments** to pinpoint counterparties with significant impacts on each XVA by country, currency and greeks. (Excel, VBA, API)
- Modeled and calculated the effects of **gap risk** on both credit and own credit risks for the portfolio of the bank.
- Developed **reporting tools** to regularly analyze gap risk, facilitating the identification of its key drivers.
- Collected and cleaned **interest rate data** for major currencies, automated data processing, and developed optimal **neural network models for yield curve prediction** with backtesting and hyperparameter tuning.

SKILLS

Programming: Python, SQL, C++, Git, VBA

AI & ML: PyTorch, Hugging Face, LLM APIs, Scikit-learn

Data & Tools: Pandas, NumPy, FastAPI, REST APIs, Docker, Power BI

Finance: Derivatives, XVA, Risk Management, Backtesting, Time Series

Languages: French, English (fluent); Arabic (fluent); Spanish (intermediate)

EDUCATION

CentraleSupélec

Sep. 2020 – Apr. 2023

Engineering Degree, Major in Mathematical Modelling & Data Science, Minor in Financial Mathematics

Paris, France

- Relevant coursework: Mathematics, Statistics, Machine Learning, Optimization, Stochastic Models, Financial Modeling, Risk Management, Derivatives, Fixed Income, Economics, Negotiation course, Data structures, Algorithms & Complexity.

Université Côte d'Azur

Sep 2017 – Apr. 2020

Bachelor's degree in Advanced Mathematics

Nice, France

- Main courses: Topology, Complex analysis, Numerical analysis, Differential calculus, Algebra & Geometry.

PERSONAL PROJECTS (AVAILABLE ON MY GITHUB)

Automatic Cold Approach – AI Outreach Platform

Sep. 2026 – Present

Python, SQLite, REST APIs, LLMs, pytest

- Goal:** A local-first, full-stack web app that systematizes cold outreach to **quant finance, data/AI consulting and applied-AI firms** – sourcing contacts, drafting personalized emails and firm-specific CV variants, and tracking every interaction end-to-end.
- Engineering & Safety:** Layered Python architecture exposed through **68 REST endpoints** with an idempotent fetch ledger (paid APIs cached, never re-spent), an append-only send ledger, and a human-in-the-loop queue (**dry-run by default**) so nothing is ever sent without an explicit decision.
- Quality:** Maintaining a **296-test suite** run against throwaway databases to protect data-provenance invariants.

Financial News Sentiment & Market Analysis Pipeline

Aug. 2026 – Present

Python, Hugging Face, FinBERT, FastAPI, PostgreSQL, Docker

- **Goal:** Building an end-to-end AI pipeline to extract sentiment from financial news, transform it into quantitative signals, and evaluate its relationship with subsequent **stock market movements**.
- **AI & Data Pipeline:** Collecting financial news and historical market data, processing and storing them in a database, and using **FinBERT** for domain-specific sentiment analysis and continuous sentiment scoring.
- **Quantitative Analysis:** Aggregating sentiment into trading signals and implementing simple **trading strategies and backtests** to evaluate predictive power, returns, and risk against baseline approaches.
- **AI Engineering:** Developing a modular Python application and progressively exposing its inference and analysis capabilities through **FastAPI**, with **PostgreSQL** and **Docker** planned for scalable data management and reproducible deployment.

Jane Street Puzzle – August 2026

Aug. 2026

Mathematics, Python, LLMs, Cursor

- **AI-Assisted Reasoning:** Using multiple frontier LLMs and **Cursor** to explore mathematical approaches, generate code and challenge proposed solutions.
- **Computational Validation:** Developing Python simulations and visualizations to test mathematical intuitions and validate analytical results.

EXTRACURRICULAR ACTIVITIES

Gap Year – Independent Travel (Southeast Asia)

Jan. 2024 – Jan. 2025

Independent Travel

Southeast Asia

- Undertook a year of independent travel across Southeast Asia, gaining cross-cultural experience and adaptability while navigating diverse environments and logistical challenges.
- Developed strong self-management, problem-solving, and communication skills through extended travel in countries including Thailand, Vietnam, Cambodia, Indonesia.
- Exposure to different cultural, economic, and social contexts strengthened intercultural awareness and resilience.